

Probability and Computing:

Exercise Sheet 5

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Question 1

Suppose we have n variables and t clauses. The variables within each clause are assumed to be distinct.

Let m_i be the number of variables required to be true, and l_i be the total number of variables in clause C_i . Clauses with $m_i > k$ variables that need to be true will not have satisfying assignments with exactly k true variables. Thus, we can ignore them and only look at clauses that need $m_i \leq k$ variables to be true.

By Theorem 4.3.5, we can use the union-of-sets sampling algorithm to create an FPRAS solution. The set U is all 2^n truth assignments. The size of the input is $t + \log |U| = t + n$.

For $1 \leq i \leq t$, the set H_i is the set of truth assignments with exactly k true variables that satisfy clause C_i . Then,

$$|H_i| = \binom{n - l_i}{k - m_i}$$

as this corresponds to setting m_i variables true, and choosing which of the remaining $(n - l_i)$ variables not in clause C_i to set to be true, which can be computed in $\text{poly}(n)$ time.

To sample uniformly from H_i , we can pick $(k - m_i)$ variables without replacement u.a.r. from the $(n - l_i)$ variables not in clause C_i .

Finally, given a truth assignment $x \in U$, we can check in poly time whether $x \in H_i$.

Question 2

By Theorem 4.3.5, we can use the union-of-sets sampling algorithm to create an FPRAS solution. The set U is all $n!$ permutations. The size of the input is $t + \log |U| = t + n \log n$.

For $1 \leq i \leq t$, the set H_i is the set of permutations that are increasing on set S_i . Then,

$$|H_i| = \binom{n}{k_i} (n - k_i)! = \frac{n!}{k_i!}$$

since it is the same as choosing k_i elements from the set $\{1, \dots, n\}$, where only 1 of the $k_i!$ permutations of k_i elements will be increasing, multiplied by the rest of the $(n - k_i)!$ permutations. This can be computed in $\text{poly}(n)$ time.

To sample uniformly from H_i , we can pick $(n - k_i)$ elements without replacement from $\{1, \dots, n\}$ u.a.r. The remaining k_i elements will be in sorted increasing order following the indices decided by S_i .

Finally, given a permutation $x \in U$, we can check in poly time whether $x \in H_i$.

Question 3

(a) Provided are 2 algorithms to part (a)

Algorithm 1 $A'(\varepsilon', \delta')$

```

 $x' \leftarrow 0$ 
for  $i = 1 \dots m$  do
     $x' \leftarrow x' + A_i(\varepsilon', 1 - (1 - \delta')^{1/m})$ 
end for
return  $x'$ 

```

Algorithm 2 $A'(\varepsilon', \delta')$

```

 $x' \leftarrow 0$ 
for  $i = 1 \dots m$  do
     $x' \leftarrow x' + A_i(\varepsilon'/m, \delta'/m)$ 
end for
return  $x'$ 

```

Algorithm 1

$$A'(\varepsilon', \delta') = \sum_{i=1}^m A_i(\varepsilon', 1 - (1 - \delta')^{1/m})$$

Note that

$$\begin{aligned} & \left(\forall i, 1 \leq i \leq m, (1 - \varepsilon')x_i \leq A_i(\varepsilon', 1 - (1 - \delta')^{1/m}) \leq (1 + \varepsilon')x_i \right) \\ & \implies \left((1 - \varepsilon') \sum_{i=1}^m x_i \leq x' \leq (1 + \varepsilon') \sum_{i=1}^m x_i \right) \end{aligned} \quad (1)$$

$$\begin{aligned} \Pr [x(1 - \varepsilon') \leq x' \leq x(1 + \varepsilon')] &= \Pr \left[(1 - \varepsilon') \sum_{i=1}^m x_i \leq x' \leq (1 + \varepsilon') \sum_{i=1}^m x_i \right] \\ &\geq \prod_{i=1}^m \Pr \left[(1 - \varepsilon')x_i \leq A_i(\varepsilon', 1 - (1 - \delta')^{1/m}) \leq (1 + \varepsilon')x_i \right] \\ &= \prod_{i=1}^m \Pr \left[1 - \varepsilon' \leq \frac{A_i(\varepsilon', 1 - (1 - \delta')^{1/m})}{x_i} \leq 1 + \varepsilon' \right] \\ &\quad \text{(by independence)} \\ &= \left((1 - \delta')^{1/m} \right)^m \\ &= 1 - \delta' \end{aligned} \quad (2)$$

Algorithm 2 Note that

$$(|x' - x| \geq \varepsilon' x') \implies \left(\exists i : \left| A_i \left(\frac{\varepsilon'}{m}, \frac{\delta'}{m} \right) - x_i \right| \geq \frac{\varepsilon' x'}{m} \right) \quad (3)$$

$$\begin{aligned} \Pr [x(1 - \varepsilon') \leq x' \leq x(1 + \varepsilon')] &= 1 - \Pr [|x' - x| \geq \varepsilon' x'] \\ &\geq 1 - \Pr \left[\exists i : \left| A_i \left(\frac{\varepsilon'}{m}, \frac{\delta'}{m} \right) - x_i \right| \geq \frac{\varepsilon' x_i}{m} \right] \\ &\geq 1 - \sum_{i=1}^m \Pr \left[\left| A_i \left(\frac{\varepsilon'}{m}, \frac{\delta'}{m} \right) - x_i \right| \geq \frac{\varepsilon' x_i}{m} \right] \quad (\text{by union bound}) \quad (4) \\ &\geq 1 - m \left(\frac{\delta'}{m} \right) \\ &= 1 - \delta' \end{aligned}$$

Algorithm 3 $A_i(\varepsilon, \delta)$

```

 $X \leftarrow 0$ 
for  $j = 1, \dots, k$  do
   $b_j \leftarrow B_i$ 
  if  $b_j \in H_i$  then
     $X \leftarrow X + 1$ 
  end if
end for
return  $X/k$ 

```

(b) Let $X_1, X_2, \dots, X_k$ be the i.i.d. indicator variables that equal 1 if $b_j \in H_i$, where b_j is obtained via B_i , and $\mu = \mathbb{E}[X_i] = |H_i|/|U_i| = x_i$. Then from Theorem 4.3.1, if $k \geq \frac{3 \log \frac{2}{\delta}}{\varepsilon^2 x_i}$

$$\Pr(|A_i(\varepsilon, \delta) - x_i| \leq \varepsilon x_i) \geq 1 - \delta$$

Thus,

$$\Pr \left[1 - \varepsilon \leq \frac{A_i(\varepsilon, \delta)}{x_i} \leq 1 + \varepsilon \right] \geq 1 - \delta$$

where $k = O(\log(1/\delta)/\varepsilon^2)$ since $\frac{1}{3} < x_i < 1$.

Algorithm 4 $A'(\varepsilon', \delta')$

```
 $X \leftarrow 1$   
for  $i = 1, \dots, m$  do  
   $b_i \leftarrow A_i(\varepsilon'/(2m), \delta'/m)$   
   $X \leftarrow X \times b_i$   
end for  
return  $X$ 
```

(c) This results in

$$A'(\varepsilon', \delta') = \prod_{i=1}^m A_i\left(\frac{\varepsilon'}{2m}, \frac{\delta'}{m}\right)$$

From part (b), we have

$$\Pr \left[\left| A_i\left(\frac{\varepsilon'}{2m}, \frac{\delta'}{m}\right) - x_i \right| \geq \frac{\varepsilon' x_i}{2m} \right] \leq \frac{\delta'}{m}$$

Then the probability that all estimates have $\left| A_i\left(\frac{\varepsilon'}{2m}, \frac{\delta'}{m}\right) - x_i \right| \leq \frac{\varepsilon' x_i}{2m}$ is

$$\begin{aligned} \Pr \left[\forall i : \left| A_i\left(\frac{\varepsilon'}{2m}, \frac{\delta'}{m}\right) - x_i \right| \leq \frac{\varepsilon' x_i}{2m} \right] &= 1 - \Pr \left[\exists i : \left| A_i\left(\frac{\varepsilon'}{2m}, \frac{\delta'}{m}\right) - x_i \right| \geq \frac{\varepsilon' x_i}{2m} \right] \\ &\geq 1 - \sum_{i=1}^m \Pr \left[\left| A_i\left(\frac{\varepsilon'}{2m}, \frac{\delta'}{m}\right) - x_i \right| \geq \frac{\varepsilon' x_i}{2m} \right] \quad (\text{by union bound}) \\ &\geq 1 - m \left(\frac{\delta'}{m} \right) \\ &= 1 - \delta' \end{aligned} \tag{5}$$

From the proof of Claim 1 in Theorem 4.4.5, we then have, using

$$\forall \varepsilon \in (0, 1), 1 + \varepsilon \leq e^\varepsilon, 1 - \varepsilon/2 \geq e^{-\varepsilon}, e^{\varepsilon/2} \leq 1 + \varepsilon,$$

$$1 - \varepsilon' \leq e^{-\varepsilon'} \leq \left(1 - \frac{\varepsilon'}{2m}\right)^m \leq \prod_{i=1}^m \frac{A_i\left(\frac{\varepsilon'}{2m}, \frac{\delta'}{m}\right)}{x_i} = \frac{A'(\varepsilon', \delta')}{x} \leq \left(1 - \frac{\varepsilon'}{2m}\right)^m \leq e^{\frac{\varepsilon'}{2}} \leq 1 + \varepsilon'.$$

Thus,

$$\Pr \left[1 - \varepsilon' \leq \frac{A'(\varepsilon', \delta')}{x} \leq 1 + \varepsilon' \right] \geq 1 - \delta'$$

From part (b) and Theorem 4.3.1, each B_i is called at least

$$k \geq \frac{3 \log \frac{2}{\delta'/m}}{(\varepsilon'/2m)^2 x_i} = O\left(\frac{m^2 \log \frac{m}{\delta'}}{\varepsilon'^2}\right)$$

times.

Question 4

Theorem 4.4.5 in the lecture notes is referred from when referencing claims and proofs mentioned.

Let $E = \{e_1, e_2, \dots, e_m\}$ denote the set of edges, and define $G_i = (V, E_i) \subseteq G$, where $E_i = \{e_1, e_2, \dots, e_i\} \subseteq E$. Let $\Omega : G \rightarrow \{\sigma\}^*$, where $\Omega(G)$ denotes the full set of proper q -colourings for graph G . Then

$$|\Omega(G)| = |\Omega(G_m)| = \frac{|\Omega(G_m)|}{|\Omega(G_{m-1})|} \cdots \frac{|\Omega(G_1)|}{|\Omega(G_0)|} \cdot |\Omega(G_0)|$$

Since G_0 has no edges, $|\Omega(G_0)| = q^n$. Considering the ratios

$$\rho_i = \frac{|\Omega(G_i)|}{|\Omega(G_{i-1})|},$$

then

$$|\Omega(G)| = q^n \prod_{i=1}^m \rho_i$$

From the lecture notes under Claim 1, we just need a $(\varepsilon/(2m), \delta'/m)$ -approximation for each ρ_i to obtain a (ε, δ') -approximation for $|\Omega(G)|$.

We then need to prove Claim 2 for q -colourings. To obtain an FPRAS, we show $\exists \delta' \in (1/2, 1)$ such that $\delta' \leq \rho_i \leq 1$.

The upper bound is easy since $\forall i, \Omega(G_i) \subseteq \Omega(G_{i-1})$. For the lower bound, we show that $\rho_i \geq 3/4$. Suppose that $e_i = (u, v)$ and denote the degree of u, v in G_i as $d_i(u), d_i(v)$ respectively. Let $d_i = \min(d_i(u), d_i(v))$. When e_i is added, only q -colourings where $\sigma_{i-1}(u) = \sigma_{i-1}(v)$ will need to be removed. Note that restricting the colouring of either u or v in order to have $\sigma_i(u) \neq \sigma_i(v)$, reducing the number of colours for the chosen vertex by 1, will imply the q -colouring will be proper for G_i . If the chosen vertex had $d+1$ choice of colours, then we can multiply $|\Omega(G_{i-1})|$ by a factor of $\frac{q-d}{q-d+1}$. To ensure the restriction, we multiply $|\Omega(G_{i-1})|$ by a factor of $\frac{q-d_i}{q-d_i+1}$, where $q-d_i$ and $q-d_i+1$ are minimum number of possible colours before and after adding edge e_i (we have for $0 < a < b < q$, $\frac{q-b}{q-b+1} < \frac{q-a}{q-a+1}$). Then we have

$$\rho_i \geq \frac{q-d_i}{q-d_i+1} \geq \frac{q-\Delta}{q-\Delta+1} \geq \frac{3}{4}$$

Thus, we have $\delta' = 3/4$, similar to the proof in the lecture notes.

ρ_i can then be estimated by drawing the appropriate number of samples from $\Omega(G_{i-1})$ using the PAUS, and calculating the fraction of them that are in $\Omega(G_i)$.

Let μ_i denote the probability that the PAUS returns a sample in $\Omega(G_i)$. Let $\hat{\rho}_i$ be the computed estimates using the PAUS.

Using the same reasoning as the lecture notes proof, fixing $\delta = \delta'/m$,

$$|\rho_i - \mu_i| \leq \frac{\varepsilon}{6m} \rho_i$$

$$|\hat{\rho}_i - \mu_i| \leq \frac{\varepsilon}{4m} \rho_i$$

$$\Pr \left[|\hat{\rho}_i - \rho_i| \leq \frac{\varepsilon}{2m} \rho_i \right] \geq 1 - \delta$$

The number of samples required is

$$\left\lceil \frac{3 \log \frac{2}{\delta}}{\left(\frac{\varepsilon}{8m}\right)^2 \mu_i} \right\rceil = O \left(\frac{m^2}{\varepsilon^2} \log \frac{1}{\delta} \right)$$

Question 5

The Markov chain M_t is **finite**, since there are $n!$ permutations. It is **irreducible**, since you can get from any permutation to any other permutation, in a sequence of swaps, similar to selection sort. It is **aperiodic**, since the same card can be chosen, so the state does not change.

Thus, it converges to the unique stationary distribution π . Define

$$(\pi_1, \pi_2, \dots, \pi_{n!}) = (1/n!, 1/n!, \dots, 1/n!)$$

Consider the system $(\pi_1, \pi_2, \dots, \pi_{n!})P = (\pi_1, \pi_2, \dots, \pi_{n!})$. Consider π_i on the RHS. There are exactly $\binom{n}{2} + 1$ states that could move to state i . These correspond to choosing any 2 cards to swap from state i , as well as self-looping choosing the exact same card. Each have probability $1/n!$ on the left. The self-loop has probability $1/n$ of transition to i , while the rest of the states have probability $2/n^2$. Thus, we obtain as desired

$$\pi_i = \frac{1}{n!} \left(\binom{n}{2} \frac{2}{n^2} + \frac{1}{n} \right) = \frac{1}{n!}$$

Let (X_t, Y_t) be a coupling of the Markov chain M_t . We choose a coupling as follows.

- Choose positions j_1, j_2 u.a.r. from $\{1, \dots, n\}$.
- Obtain X_{t+1} from X_t by swapping the card at position j_1 with card at position j_2 . Call the card at position j_2 card C .
- Obtain Y_{t+1} from Y_t by swapping the card at position j_1 with card C .

We show that Y_t is a faithful copy of the original chain. Consider any position k_1, k_2 . Let C' be the card in position k_2 in Y_t . Let j_2 be the position of C' in X_t , and let $j_1 = k_1$. The probability that j_1, j_2 are chosen are each $1/n$, so this is the probability that position k_1, k_2 are chosen in the transition from Y_t to Y_{t+1} .

Note that when a card C is moved to position j_1 in (X_t) and (Y_t) it thereafter occupies the same position in both decks. The probability that a given $C \in \{1, \dots, n\}$ is chosen in each transition is $1/n$.

Thus, we have a coupon collector problem. As in the lecture notes for the card shuffling example, the probability that the chain has not coupled in t steps is at most $\exp(-\frac{t}{n} + \log n)$. Choosing $t = n(\log(1/\varepsilon) + \log n)$ gives that this probability is at most ε . By coupling lemma, the mixing time is at most

$$n(\log(1/\varepsilon) + \log n)$$

Question 6

(a)

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 \end{pmatrix}$$

(b) The Markov chain is **finite**, since there are n states. It is **irreducible**, since it can get from any state i to j by going to 0 with a tails, and tossing head until reaching j . It is **aperiodic**, since it can use the self-loop on state n to end up returning to each state with even or odd steps.

Thus, there is a unique stationary distribution π . Define

$$\pi = (1/2, 1/2^2, 1/2^3, \dots, 1/2^{n-1}, 1/2^n, 1/2^n)$$

so $\pi_i = 1/2^{i+1}$ for $0 \leq i < n$. Consider π_i .

$$\forall i, 0 < i < n, \pi_i = \frac{1}{2}\pi_{i-1} = \frac{1}{2^{i+1}}$$

$$\pi_0 = \frac{1}{2} \sum_{i=0}^n \pi_i = \frac{1}{2}$$

$$\pi_n = \frac{1}{2}(\pi_{n-1} + \pi_n) = \frac{1}{2^n}$$

Then $\pi = \pi P$. The chosen π is the unique stationary distribution.

(c) We define a coupling (X_t, Y_t) by having (X_t) and (Y_t) behave like independent copies of the Markov chain if $X_t \neq Y_t$, and otherwise have them behave identically. Note that

$$\forall i, j, i \neq j, \Pr[(X_{t+1}, Y_{t+1}) = (0, 0) | (X_t, Y_t) = (i, j)] = 1/4$$

Then

$$\Pr[X_{t+1} = Y_{t+1} | X_t \neq Y_t] \geq 1/4$$

The probability of not being equal after t steps is at most $(3/4)^t$. Set $\varepsilon \geq (3/4)^t$, so $t \geq \log_{4/3}(1/\varepsilon)$ and

$$\Pr[X_t \neq Y_t] \leq \varepsilon$$

By the coupling lemma, the mixing time is bounded by $O(\log(1/\varepsilon))$.

(d) Note that this Markov chain is also finite, irreducible, and aperiodic, since it is the time-reversal. Thus, it has a unique stationary distribution. In fact, its stationary distribution is exactly the same as before.

$$\hat{\pi} = \pi = (1/2, 1/2^2, 1/2^3, \dots, 1/2^{n-1}, 1/2^n, 1/2^n)$$

Observe that

- $\forall i < n, \hat{P}_{0,i} = \frac{1}{2^{i+1}}$
- $0 \leq i < n-1, \hat{P}_{i+1,i} = 1$
- $\hat{P}_{n,n-1} = \hat{P}_{n,n} = \frac{1}{2}$

Analysing the $\hat{\pi}_i$,

$$\forall i, 0 < i < n, \hat{\pi}_i = \hat{\pi}_{i+1} + \frac{1}{2^{i+1}} \hat{\pi}_0 = \frac{2}{2^{i+2}} = \frac{1}{2^{i+1}} = \hat{\pi}_i$$

$$\hat{\pi}_0 = \frac{1}{2}(\hat{\pi}_0 + \hat{\pi}_1) = \frac{1}{2} = \hat{\pi}_0$$

$$\hat{\pi}_n = \frac{1}{2} \hat{\pi}_n + \frac{1}{2^n} \hat{\pi}_0 = \frac{2}{2^{n+1}} = \frac{1}{2^n} = \hat{\pi}_n$$

Then $\hat{\pi} = \hat{\pi} \hat{P}$.

Note that since for $0 \leq i < n-1, \hat{P}_{i+1,i} = 1$ then for $0 < i < n, \hat{P}_{i,0}^i = 1$.

Define $\vec{1}_i = (0, \dots, 1, \dots, 0)$ where it is equal to 1 at index i , and 0 otherwise.

Consider an arbitrary $i, 0 \leq i < n$, such that $\hat{\pi}_B^0 = \vec{1}_i$. Then since $\hat{P}_{i,0}^i = 1$, we have $\hat{\pi}_B^i = \vec{1}_0$. Then we must have $\hat{\pi}_B^{i+1} = \hat{\pi}$, where clearly $i+1 \leq n$. Since it is in a stationary distribution, it will stay there so $\hat{\pi}_B^n = \hat{\pi}$.

For the case $i = n$ where $\hat{\pi}_B^0 = \vec{1}_n$, note that in each time step $t \leq n$, for $i < n$,

$$(i+t \geq n) \implies \hat{\pi}_{B,i}^t = \frac{1}{2^{i+1+t-n}}$$

and $\hat{\pi}_{B,n}^t = \frac{1}{2^t}$.

Then at time step $t = n$, we have

$$\begin{aligned} \hat{\pi}_{B,i}^n &= \frac{1}{2^{i+1}} \\ \hat{\pi}_{B,n}^n &= \frac{1}{2^n} \end{aligned}$$

Thus, $\hat{\pi}^n = \hat{\pi}$. We have

$$\forall i, \hat{\pi}_B^0 = \vec{1}_i \implies \hat{\pi}_B^n = \hat{\pi}$$

Next, we note that for any distribution $\hat{\pi}_D^0 = (p_1, p_2, \dots, p_n)$, it can be expressed in terms of $\vec{1}_i$ as

$$\hat{\pi}_D^0 = (p_0, p_1, \dots, p_n) = \sum_{i=0}^n p_i \vec{1}_i$$

where $\sum_{i=0}^n p_i = 1$. Since after $t = n$ steps, we know that $\vec{l}_i$ will be in the stationary distribution $\hat{\pi}$,

$$\hat{\pi}_D^n = \sum_{i=0}^n p_i \hat{\pi} = \hat{\pi}$$

Then the mixing time is at most n for any distribution of $\hat{\pi}_D$.